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A Research Project on

Smart Serve: The Effect of Digital Job Application to the Employment of The Applicant in Municipality of Polomolok

As partial requirement for the Subject

IT ELEC 4

Advanced Human Computer Interaction

By:

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February 2025

1.0 INTRODUCTION

1.1 Background and Context

Applying for a job has never been particularly smooth, cases it might take up to a year or even months. You print your résumé, place it in a brown envelope, travel to the municipal hall, and stand in line hoping someone files your documents correctly. In the Municipality of Polomolok scenario is not a “might” but is actually happening. The hiring process still depends largely on paper submissions. Résumés. sit in stacks on desks. Cover letters disappear somewhere between departments. From the applicant’s side, the experience can feel uncertain and exhausting. Not only that but the HR staff and the employer does not have it easy either, they have to file those documents manually and manage it manually, that might even take hours, sometimes days, trying to keep everything organized. For a small to medium. local government unit, that kind of system quietly drains time and energy.

And the cost is not just inconvenience. When applications are handled manually, Capable applicants sometimes get overlooked since their documents gets buried. And That is a problem. we could no longer ignore. We began to ask ourselves whether a practical, realistic solution was within reach. The solution we came up with is Instead of adding more paperwork or complicated process, we thought of a capable web-based system made specifically. for our municipality. A platform that applicants could access on their smart device without being good at using technology. A system that would allow HR personnel to focus. on evaluating qualifications rather spend hours arranging documents. We chose web technologies precisely because they are widely accessible and familiar. At the same time, we were careful. to prioritize security and simplicity, recognizing that any digital shift in a government office must be both usable and responsible.

1.2 Research Problem

The thing is the way things work now at the Municipality of Polomolok. When someone tries to apply for a job, here is what typically transpire:

1. They print multiple copies of their resume, wasting paper and money.
2. They go to the municipal office and submit documents which is a hassle.
3. HR staff manually log each applicants, application in spreadsheets.

4. The same person sometimes applies multiple times without realizing it which would cause redundancy of data.
5. There is no way for applicants to check if their application was received or reviewed unless they physically inquire.
6. Sensitive personal information sits in filing cabinets that anyone could access

These are not just minor inconveniences. They represent real barriers to effective hiring. Good candidates get discouraged by the complicated process. HR staff gets tired processing files. And the municipality misses out on potentially talented individuals that could serve the community better. This research aims to solve these problems through thoughtful system design.

1.3 Research Questions and Objectives

What we need to figure out:

1. Can a digital system actually speed up hiring without making things more complicated?
2. What do applicants and HR staff really need from a job platform?
3. How do we keep personal data safe while still making the system user friendly?

1.4 Justification and Significance

Why does this matter? For starters, it is 2025. Paper-based hiring systems feel outdated, and they are. But beyond that, there is a real opportunity here to make government service more accessible to everyone. When applying for a municipal job is as easy as ordering food online, more people will apply. That means a bigger talent pool and ultimately better public service. For job seekers or applicants, the ones who have grown up with technologies, this system removes barriers. They can apply from anywhere as long as they have a device, and get updates without making phone calls or visiting offices. For our HR team, it means less time on paperwork and more time finding the right people for the job.

2.0 LITERATURE REVIEW

2.1 Overview of HCI Theories and Models

Related Literature

Usability and Human-Computer Interaction (HCI)

Issa and Isaias provide a comprehensive overview of how HCI principles shapes modern system designs. Their work states that for any digital system to be effective, it must focus on the user's need from the very beginning of the developments process. What stands out in their discussion is on how they use usability to sustainability showing that systems people actually want to use are the one that survive long-terms. For our Job Finder System, this means we cannot just build something functional that would work but, we also need to build something people will actually enjoy using for long term use. The authors stress that involving users early through prototyping and feedback loops is not optionalit is essential. Their framework of usability gives us concrete criteria to evaluate our interface design, from visibility of system status to error prevention.

Human Resources Information Systems: A Guide for Public Managers

This textbook is basically the bible for anyone building HR systems for government. Benavides and his co-authors bring decades of real-world experience from municipal government settings, which makes their advice incredibly relevant to our project. They analyze everything from system selection to application of it, with special attention to the challenges public organization face like dealing with civil service rule, union requirement, and political oversights. What I found most useful relevant is their discussions of how HRIS can transform administrative effectivness while maintaining compliances. They also provide practical framework for evaluating different system architectures and vendor solution. For municipalities like Polomolok, their guidance on phased implementation and change management is particularly valuable.

Trustworthiness in E-Recruitment Systems

Ogunniye and colleagues tackled something that does not get enough attentions in system designs, trust. Their researches show that even the most technically perfect job system and platform will fail if user do not trust it which shows reliability and trust is needed. They identified several factor that builds or destroy trust in e-recruitment systems transparency about how data is used, clear communication throughout the application process, and professional design that signals that it is not fraud. What surprised me was how much visual design matters users make trust

judgments in seconds based on how a site looks. Their findings about explanation-enhanced systems are especially relevant - when users understand why certain information is requested or why an application status changed, trust increases significantly. For our municipality system, this research tells us we need to be extra clear about data handling and provide explanations for every step.

Shared Values of E-Recruitment Portals

Rahman and Patra's study digs into what actually makes job seekers choose to use (or abandon) online job platforms. They surveyed hundreds of job seekers and found that trustworthiness, user experience, and performance expectancy are the biggest factors influencing whether someone will actually complete an application. Their model shows these factors work together - a fast-loading site with poor design will still fail, and a beautiful site that does not work well will frustrate users. What makes this research practical is that they showed us a specific metric for each factor, giving us benchmarks to aim for. For example, they found out that user expect pages to load in under 3 seconds and forms to be complete in 15 minutes. Their work on subjective norm - how much users care about what others think of the platform - also suggests that building a good reputation in the community matters.

System Usability Scale and Technology Acceptance Model

Pal and Vanijja's research gives us solid methodological tools for evaluating our system. They combined two well-established frameworks - the System Usability Scale (SUS) and the Technology Acceptance Model (TAM) - to assess Microsoft Teams during the pandemic. Their approach is directly applicable to our project because they were evaluating a system that had to work for diverse users with varying technical skills, which is exactly our situation at the municipality. The SUS gives us a standardized 10-question survey that produces a single usability score we can compare against industry benchmarks. TAM helps us understand the factors that drive adoption - perceived usefulness and perceived ease of use. Their finding that these two factors together predict over 70% of user acceptance gives us clear targets: our system must be both useful and easy to use, not just one or the other.

Related Studies

Accessibility and Usability of Online Job Applications for Screen Reader Users

This study hits close to home because accessibility is often an afterthought in government systems. Reuschel and his team tested 30 Fortune 500 company job applications with actual screen

reader users and found that every single one had accessibility barriers. The most common problems were improperly labeled form fields, date pickers that screen readers could not navigate, and error messages that were not announced. What makes this study powerful is that they had real users with visual impairments do the testing, not just automated tools. The results were sobering - many applicants could not complete applications independently. For our Job Finder System, this research is a wake-up call. We need to build accessibility from the start not add it later. The authors provide specific WCAG 2.1 guidelines to follow and testing ways that include actual user with disabilities.

HR Information System Effectiveness in Tanzanian Local Government

Matimbwa and colleagues studied HRIS implementation in local government authorities in Tanzania, which shares many similarities with Philippine municipalities. Their mixed-methods research surveyed HR staff and did interviews across multiple local governments to understand what makes the systems actually work. They found that the system is effective, depends heavily on three factors: reliable infrastructure, user training, and management support. What is particularly relevant is their finding that even technically good systems failed when users were not properly trained. They also discovered that local governments need systems designed for their specific context - generic enterprise solutions often required too much customization. Their recommendation for phased implementation, starting with core features and adding complexity over time, is something we are considering for Polomolok. The study provides a practical framework for measuring HRIS effectiveness that we can adapt.

Digital Human Resources Management in Bangladesh Government Health Sector

Bhattacharyya and her team analyzed the implementation of a digital HRM system in Bangladesh's government health sector, which faced difficulties remarkably similar to what we expect in Polomolok. Their policy content examined documents, interviewed stakeholders, and tracked the process over time. They found that successful digital transformation in government needs more than just good technology it needs policy alignment, leadership commitment, and change management. One of their key findings was that resistance from staff was the biggest

barrier, not technical issues. They also discovered that having champions at multiple levels - from top management to frontline staff - made a huge difference in adoption. Their framework for analyzing readiness and their recommendation for addressing resistance are directly usable to our project. The study shows that government HR systems must balance efficiency with transparency and accountability.

User Experience Analysis of AI-Based Interview Platform

This very recent study from Indonesia examines how job seekers experience AI-powered interview platforms, which represents the improvement of recruitment technology. Anbiya and colleagues used both quantitative (SUS scores) and qualitative (think-aloud protocols) methods to understand users. They found that while AI features can improve efficiency, they also create anxiety for user who don’t understands how decision are made. The sessions revealed that users need clear explanations of what the AI is doing. Their methodology - combining standardized surveys with observational techniques - gives us a model for our own user testing. The study also found that younger users adapted more quickly to AI features, suggesting we need to design for varying comfort levels with technology. For our system, this research suggests we should be cautious about AI features and prioritize transparency if we include them.

Digital HR Systems Adoption in Public Institutions

Jimoh's 2025 study is one of the most recent and relevant pieces of research on our topic. He investigated how digital HR system adoption affects administrative efficiency in public institutions, with a specific focus on the mediating role of employee digital literacy. His quantitative study of over 300 public sector employees found that digital HR systems do improve efficiency, but only when employees have adequate digital skills. This is a crucial finding for Polomolok - it means our system design must account for varying levels of digital literacy among staff. Jimoh also found that training programs significantly improved adoption rates and system effectiveness. His structural equation model shows the relationships between system quality, user satisfaction, and actual use, giving us a theoretical foundation for our design decisions. The study recommends that public institutions assess digital literacy before implementation and plan training accordingly.

Synthesis of Literature and Studies

Looking at all this research together, some clear themes emerge that should guide our Job Finder System design. First, usability is not optional - it is the foundation everything else builds

on. Issa and Isaias, Pal and Vanijja, and Reuschel's team all emphasize that systems must be designed with users from the start, not tested on them at the end. Second, trust matters enormously. Ogunniye's work on trustworthiness and Rahman's findings about shared values show that technical functionality alone is not enough - users need to feel confident in the system. Third, context is everything. Matimbwa's study in Tanzania, Bhattacharyya's work in Bangladesh, and Jimoh's research all show that government systems face unique challenges that generic solutions do not address well.

The related studies also reveal practical lessons. Reuschel's accessibility testing shows we must include users with disabilities from the beginning. Matimbwa and Bhattacharyya both found that training and change management are as important as the technology itself. Anbiya's AI research suggests we should be thoughtful about advanced features and prioritize transparency. Jimoh's finding about digital literacy as a mediator tells us we need to assess our users' skills and plan training accordingly.

For our specific context at the Municipality of Polomolok, this body of research points to several key design principles: keep interfaces simple and intuitive, build accessibility in from the start, provide clear explanations at every step, plan for comprehensive training, and design for the specific needs of municipal government hiring. The literature gives us both the theoretical foundation and practical guidance to build a system that actually works for everyone involved.

3.0 METHODOLOGY

3.1 Research Design

This study adopts a **quantitative & developmental research design**, combining statistical investigation with system development. The goal is not only to measure existing recruitment challenge in the municipality of polomolok but to design and develop, and evaluate a functional web recruitment system grounded in empirical data. Quantitative methods were selected because the project requires measurable evidences. Rather than relying on impressions or informal feedback that is unreliable, the study collects numerical data to identify patterns in current hiring practice, assess user need, and evaluate system performance after application. At the same time, the developmental components ensure that finding are translated into a working prototype that directly respond to the data gathered by the researcher. process will looks like this:

3.2 Participants

The study involves four primary participant groups who are directly involved or connected to the municipal recruitment processes. First are the job seekers or applicants, ranging from fresh grads to experienced professionals, whose varying levels of technological familiarity must be considered to ensure the system remain accessible. Next are the HR personnel who manage application daily and possess practical insight into procedural inefficiencies and required documentation. Department heads are also included, as they are responsible for final hiring decision and require structured report and comparative dashboards to evaluate candidates effectively.

3.3 Data Collection

Data are gathered through multiple methods to ensure depth of information. Surveys are distributed to capture quantifiable data on user needs, feature preference, and the current state in the current recruitment process. To complement these numerical finding, interviews are conducted with HR personnel and department heads to gain detailed insight into workflow challenges and decision-making practices. We then proceed with Usability testing, during which participants are observed while completing specific task within the prototype system. Following system deployment, we will monitor how often they use or the patterns, such as login likeliness or frequency

and feature engagements, with participant consent of course. And we will be integrating feedback forms ensuring continuous data collection beyond the initial testing phase.

3.4 Data Analysis

Collected data are analyzed using both statistical and thematic techniques appropriate for a quantitative developmental framework. Survey responses are processed through descriptive and inferential statistical methods, majority preferences, and areas of divergence among user group. Usability testing results are evaluated using measurable indicator such as task completion rate, time on task, and error frequency, allowing pattern of performance to be identified across the participants. System impact and overall user satisfactions are assessed using standardized instruments, including the SUS scale, to produce scores that reflect perceived usability and effectiveness.

3.5 Ethical Considerations

Ethical standards are strictly observed throughout the research process due to the involvement of personal and the potential sensitive information that may leak. All participants provide informed consent before they can participate and are clearly briefed on the objectives, procedures, and scope. Anonymity is maintained by removing personal information from datasets whenever feasible, and collected data are stored securely with restricted access to authorized personnel only. Transparency is used by informing participants about how their data will be used, analyzed, and retained. Moreover, participants have the right to withdraw from the study at any stage without penalty or obligation to provide justification, ensuring that participation remains entirely voluntary.

Features and Functionalities

So, what exactly are we building? Here are the core features that address the problems we identified:

1. Centralized Application Management - One place for all job postings and applications.
2. User Management - Different access levels for applicants, HR staff, and admins.
3. Workflow Automation - Automatic status updates and notifications.
4. Reporting Dashboard - Visual summaries of application metrics.
5. Duplicate Detection - Smart checks to prevent the same person from applying multiple times to the same position.
6. Skill & Profile Matching – Automatically match the applicants' skills on his profile to match the job.
7. Notification (Email) – Sends notification about the applicants.

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